

# Inside A Compás

A flamenco metronome that is a web app wearing a native shell. Thirty-four packages build it; roughly half of them reach the user. Here is what each one is for.

**Quasar** 2.28.0    **Vue** 3.5.42    **Capacitor** 7.6.8    **Tone.js** 15.1.22    **TypeScript** 5.9.3

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Every pixel you see is a web page. It runs in a browser tab on `acompas.org`, inside a `WKWebView` on iOS, and inside Chromium on the desktop — the same code in all three. What changes is the container.

That is worth holding onto, because it explains the shape of the dependency list: one set of packages builds a web application, a second wraps it in something installable, and a third never ships at all.

# What you see

The interface, its language, and its motion.

SHIPS TO USERS

**vue** 3.5.42

The framework everything else is built on. You describe what the screen should look like for a given state; Vue keeps the DOM matching it.

**quasar** 2.28.0

A component library and design system for Vue — buttons, dialogs, drawers, toggles, dark mode, the responsive grid. It is why the app looks consistent without hand-written CSS for every control.

## IN THIS APP

Three of its plugins are switched on: `Notify` (the toast that tells you to select an instrument), `Loading`, and `Dialog`.

**pug** 3.0.4

An indentation-based template language used instead of HTML. Every `.vue` file here declares `<template lang="pug">`, which is why the markup has no closing tags.

## IN THIS APP

Compiles to plain HTML at build time, so it costs the user nothing.

**vue-router** 5.3.0

Maps URLs to screens. Each page has its own address.

## IN THIS APP

Uses real URLs on the web and hash URLs (`#/flamenco/solea`) in the native builds, because there is no server to ask for a path when the page is loaded from inside the app bundle.

#### **vue-i18n** 9.14.5

Translation. Text lives in per-language files rather than in components.

##### **IN THIS APP**

Ten languages, including Arabic and Persian, which read right to left. There is deliberately no i18n build plugin: the messages here are `.ts` modules, not the JSON/YAML resources such plugins precompile, and the current one defaults to a vue-i18n build with no message compiler — which silently emptied every translated control while the build still succeeded.

Translation. Text lives in per-language files rather than in components.

##### **IN THIS APP**

Ten languages, including Arabic and Persian, which read right to left.

`@intlify/vite-plugin-vue-i18n` compiles those files at build time so translation lookups are fast at runtime.

#### **animejs** 3.2.2

A small animation library. Given a target and a property, it interpolates over time.

##### **IN THIS APP**

The pulse on each dot as its beat lands — scale up, ease back down.

#### **@quasar/extras** 1.18.0

Icon sets and fonts packaged for Quasar. Only `mdi-v7` (Material Design Icons) is enabled; the other five are commented out so they are never downloaded.

## vue-global-events

3.0.1

Lets a component listen for keyboard events on the whole window without manually adding and removing listeners. Powers the keyboard shortcuts.

## What it remembers

Shared state, and the parts that survive a restart.

SHIPS TO USERS

### pinia 2.3.1

The state store. Rather than passing data down through components, everything reads from a central place and updates flow outward.

#### IN THIS APP

Two stores. `patterns` holds the selected palo, tempo, instruments and beat events; `session` holds preferences like dark mode and the audio-visual offset.

### @vueuse/core 10.11.1

A large collection of small Vue utilities.

#### IN THIS APP

Almost entirely one function: `useStorage`, which mirrors a value into `localStorage` automatically. It is why your tempo, language and chosen pattern are still there next time.

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## What makes the sound

The hard part, and the reason a metronome is not just a timer.

SHIPS TO USERS

**tone** 15.1.22

A framework over the browser's Web Audio API. Its value is scheduling: JavaScript timers drift audibly, so Tone schedules notes against the audio hardware clock instead, looking ahead and queueing events before they are due.

### IN THIS APP

`Transport` is the clock that tempo and swing act on. `Sequence` plays one pattern array per instrument. `Player` holds each decoded sample, `Channel` mixes them, `Reverb` adds room, and `Draw` schedules the dot animations against the same clock so sight and sound stay together.

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## The written material

Help pages, palo descriptions, the changelog.

SHIPS TO USERS

**marked** 14.1.4

Turns Markdown into HTML. The help text and palo documentation are authored as Markdown inside the translation files.

**dompurify** 3.4.14

Strips anything dangerous out of HTML before it is inserted into the page. Standard practice whenever generated HTML is injected — it closes off cross-site scripting.

### IN THIS APP

Paired with `marked` : parse, then sanitize, then render. Some of that text is fetched from Wikipedia, so it is not all locally authored.

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## The native shell

What turns a web page into something you install.

SHIPS TO USERS

### **@capacitor/core** 7.6.8

Wraps the web app in a real native application: a Swift or Kotlin project containing a full-screen web view, plus a bridge that lets JavaScript call native code. The web assets are copied inside the app bundle and served through a custom URL scheme, so the app works with no network at all.

#### IN THIS APP

That scheme handler is also where the iOS audio bug lived — it answers requests for media files in a way that `fetch()` reads as a failure.

### **@capacitor/app · splash-screen · status-bar**

7.1.2 / 7.0.5 / 7.0.6

Small official plugins exposing native behaviour to JavaScript: lifecycle events such as pause and resume, the launch splash, and the colour and style of the status bar.

### **@capacitor-community/keep-awake** 7.1.0

Stops the screen sleeping. For a metronome propped on a music stand, this is not a nicety.

### **electron** 31.7.7

The desktop equivalent: bundles Chromium and Node.js into a Mac or Linux application. Heavier than Capacitor — the browser engine ships with the app rather than being borrowed from the operating system.

#### IN THIS APP

`@electron/remote` lets the page reach a few main-process APIs;  
`electron-builder` produces the installers.

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## What builds it

Runs on your machine. None of this reaches a user.

NEVER SHIPS

### **@quasar/app-vite** 3.8.1

The build system and CLI. It owns `quasar dev` and `quasar build`, wires Vue, the router, the store and Quasar together, and knows how to target each platform — the reason `-m capacitor -T ios` means anything.

#### WORTH KNOWING

It carries its own Vite (8.2.2 here) rather than taking it as a dependency. The project moved v1 → v2 → v3 in one sitting; each hop broke the config differently, and both of the interesting failures compiled cleanly before breaking at runtime.

### **typescript** 5.9.3

JavaScript with type annotations, checked at build time and then erased. The pattern and sound data are typed, which is what makes a malformed sequence a build error rather than silence at runtime.

### **eslint · prettier** 8.57.1 / 3.9.6

ESLint catches suspect code; Prettier formats it. `eslint-config-prettier` switches off the ESLint rules that would argue with Prettier, and `eslint-plugin-vue` teaches ESLint about `.vue` files.

### **autoprefixer** 10.5.4

Adds vendor prefixes to CSS for the browsers listed in the build target, so you write the standard property once.

### **electron-builder** 24.13.3

Packages the Electron app into a `.dmg` or Linux package, handling signing and metadata. Selected in `quasar.config.js` via `bundler: 'builder'`.

### **@capacitor/cli** 7.6.8

Copies built web assets into the native projects and keeps their plugin lists in sync. Quasar calls it for you during a Capacitor build.





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## Declared but not carrying weight

A grep for package names finds nothing for a third of the dependency list. Some of those were genuinely dead and have been removed; the rest are invoked as tooling or referenced by a name that is not the package path.

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### PACKAGE

### WHY IT IS HERE

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removed

Six packages were dropped on 2026-08-30: `pug-plain-loader` (a webpack loader in a Vite project), `electron-packager` (the config selects `builder`; deprecated upstream), `@types/dompurify` (DOMPurify 3 ships its own types), `@vue/devtools` (app-vite v3 uses `vite-plugin-vue-devtools`, installed on demand), `@capacitor/assets` (icons come from `yarn icons`) and `@vue/language-plugin-pug` (never wired into `vueCompilerOptions`).

@types/animejs

Types for anime.js, which ships none of its own — genuinely load-bearing despite nothing importing it by name. Kept.

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## PACKAGE

## WHY IT IS HERE

@quasar/extras

Named in `quasar.config.js` as `'mdi-v7'` rather than by package path, so a search for the package name misses it.

eslint · typescript · autoprefixer · electron-builder · @capacitor/cli

Invoked as tooling by the build or by ESLint's own config; never imported from source.

Versions shown are what is installed in `node_modules`, which can sit ahead of the range declared in `package.json` — `dompurify` is declared as `^3.1.6` and installed at `3.2.6`.

The audio samples are not a package. They live in `public/audio` as 321 recordings in five formats, generated from WAV masters by `format_audio.py`, and account for roughly 99 MB of the iOS app.